

SUMMARY

Curious and quick to learn, I'm committed to mastering C++ and strong software design habits. I write code that's clean, easy to understand, and backed by basic unit tests, using repeatable workflows to ensure reliability. Right now, I'm building a portfolio of small, well-structured projects including game development prototypes to prove my skills in action. I'm eager to grow into Linux, tooling, and gameplay systems development, where I can contribute to impactful solutions while sharpening my craft.

TECHNICAL SKILLS

C++(intermediate) | OOP / OOD | STL | Boost(familiar) |
CMake(basic) | Unit testing (learning) | Perforce | Git |
UMTS/LTE (basic) | Linux(basics) | UE5 | Unity

PROFESSIONAL SKILLS

Problem solving | Debugging | Team collaboration | Jira
Agile teamwork & sprint delivery | Time management |
Technical communication | Project ownership

PROJECTS

TG 3SIXO - HackYeah Hackathon Project — Real-Time Traffic Simulation Backend

Sep 2025 - Oct 2025

C++ | JS/HTML | CMake | FastAPI

- Developed a predictive traffic engine combining live transit data, user reports, and AI forecasting.
- Implemented Dijkstra-based routing, real-time SSE updates, and delay prediction via TransitDNA.
- Built a user-facing map UI showing baseline vs. disrupted routes, ETA changes, and proactive alerts.
- Integrated calendar conflict detection, dispatcher API, and live GTFS data for realistic scenarios.

C++ Game Engine — Custom rendering pipeline (in progress).

Nov 2025 - Current

C++17 | OpenGL | GLSL | GLM | GLFW | ImGui

- Developing a modular game engine featuring a renderer, scene manager, shader system, and simple ECS.
- Implemented vertex/element buffers, shader abstraction, perspective projection, entity rendering, and transform logic.
- Rendering textured 3D objects (triangle/cube) with camera controls and UI integration using ImGui.

Gnome Splash - Local Multiplayer Game

Mar 2025 - Apr 2025

C++ | Unreal Engine

- Led multiplayer gameplay programming for a local split-screen PvP game, created damage, respawn, and player spawning systems.
- Developed UI and HUD features, like health bars, cooldowns, and menus.
- Optimized player experience by integrating enhanced input, dynamic spawn locations, and smooth match flow from menus to gameplay.

Additional Projects

EDUCATION

Bachelor of Engineering in Computer Science

Aug 2020 - Jun 2024

Chandigarh University

Diploma (Game Programming)

Sep 2024 - Dec 2026

FutureGames

ADDITIONAL INFORMATION

- **Languages:** English (Fluent) , Hindi (Native) , Gujarati (Native)
- **Certifications:** Intermediate OOPS for Unity Games , Object-Oriented Data Structures in C++ , Cloud Virtualization , Containers and APIs
- **Awards/Activities:** Organized a Hackathon in 2022 , Participated In Hackyeah2025

References

Jeremiasz Kacprzak

Mentor/Programmer at The Astronauts

Phone: +48 782 827 105

Email: kacprzakjeremiasz@gmail.com

Maria Glowacka

Senior Franchise Manger - CD Projekt Red

Phone: +48 502 608 571

Email: majkaglowacka@icloud.com

Piotr Matecki

Head of FutureGames Warsaw

Phone: +48 799 094 740

Email: piotr.matecki@futuregames.se